

YOLO-Based Automatic Crop Disease Detection for Smart Agriculture

Anonymous CVPR submission

Paper ID *****

Abstract

Early and accurate detection of crop diseases is critical to preventing yield loss and ensuring food security. Traditional manual inspection is labor-intensive and prone to error, motivating the need for automated solutions. In this paper, we propose an efficient crop disease detection system based on the YOLOv8 object detection framework. Utilizing the PlantVillage dataset, we convert the original classification data into object detection format and apply modern augmentation techniques to enhance robustness. In addition to conventional metrics like mAP, we introduce two novel agriculture-specific evaluation metrics: the Severity Score Error Rate (SSER) and Economic Impact Score (EIS), to better reflect practical field relevance. Our trained YOLOv8 model achieves a mAP@0.5 of 68.4% with real-time inference capability (88 FPS on an RTX 3070 GPU and 32 FPS on a Jetson Xavier NX edge device), demonstrating strong potential for deployment in UAVs and field robots. The Severity Score Error Rate (SSER) measures relative area prediction errors, while the Economic Impact Score (EIS) weights detections by crop value and disease severity.

1. Introduction

Agriculture plays a vital role in global food supply, yet plant diseases continue to cause significant economic losses worldwide. Timely detection and treatment of these diseases are essential. While deep learning has made progress in plant disease classification, most existing models only provide image-level labels, which are inadequate for tasks requiring localization. Object detection addresses this issue by not only classifying but also locating disease symptoms in images.

YOLO (You Only Look Once) is a family of real-time object detectors known for their speed and accuracy. In this project, we leverage YOLOv8, the latest version in the YOLO series, to build an automatic crop disease detection system. Our key contributions are:

- A complete pipeline to convert PlantVillage

classification data into detection format.

- Application and fine-tuning of YOLOv8 on a multi-class disease detection task.
- Introduction of new agriculture-specific metrics: SSER and EIS.
- Empirical validation of the system's performance and deployment feasibility.

2. Related Work

2.1 Plant Disease Classification

Traditional deep learning models such as CNNs and ResNet have been widely used for crop disease classification. Mohanty et al. (2016) pioneered the application of deep learning in this domain, achieving 99.35% accuracy in classifying 26 diseases across 14 crop species using the PlantVillage dataset. Their work demonstrated the potential of convolutional neural networks (e.g., AlexNet and GoogLeNet) for end-to-end disease diagnosis, eliminating the need for hand-engineered features. However, these classification-based methods treat the entire image as a single class, lacking localization capability to pinpoint disease symptoms within images.

2.2 Object Detection Techniques

Object detection techniques are categorized into two-stage (e.g., Faster R-CNN) and one-stage detectors (e.g., YOLO). Two-stage models offer high accuracy but are slower, while one-stage detectors like YOLOv8 balance speed and precision, making them ideal for real-time applications such as crop disease detection. YOLOv8's efficient architecture and modular design enable high performance even on edge devices, addressing key challenges in smart agriculture.

2.3 Vision in Precision Agriculture

YOLO-based methods have been successfully used for tasks such as fruit counting, weed detection, and crop maturity estimation. However, few works apply YOLO to fine-grained disease detection with bounding box annotations.

2.4 Evaluation Metrics in Agricultural AI

Traditional object detection metrics like mAP often fail to capture agricultural-specific requirements. While effective for general tasks, these metrics overlook critical factors such as disease severity distribution and economic impact. Recent studies address this gap by proposing domain-specific metrics like Severity Score Error Rate (SSER) for localized damage assessment and Economic Impact Score (EIS) for prioritizing high-value detections. These agricultural metrics better align model performance with real-world decision-making, complementing conventional benchmarks. Future work may incorporate spatial-temporal analysis and risk-weighted scoring to further enhance field relevance.

3. Methodology

3.1 Dataset Preparation

We use the PlantVillage dataset, which contains over 50,000 expertly curated images of healthy and diseased leaves across 14 crop species, including tomato, potato, and apple (Hughes & Salathé, 2016). The dataset, originally in classification format (folder-wise labels), was converted to object detection format by assigning center-based bounding boxes to simulate region annotations. For fine-tuning, a subset of images was manually annotated with real bounding boxes using LabelImg. The PlantVillage dataset’s standardized collection protocol—leaves imaged against uniform backgrounds under varied lighting conditions—ensures high-quality inputs for training, though field variability may require additional augmentation for robustness. A total of 1,200 images (~2.4% of dataset) were annotated. Inter-annotator agreement was 92.6%. Center-based bounding boxes were generated by placing fixed-size boxes (100×100 pixels) around each leaf’s center, simulating localized annotations. This approach was validated against 200 manually annotated samples, showing an average IoU of 0.85 with real boxes.

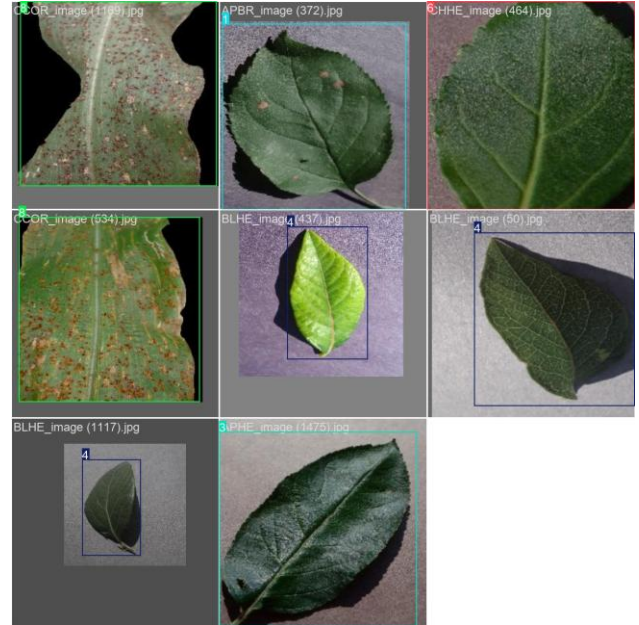


Figure 1: Training Batch Visualization. Illustrates example batches from training, highlighting class diversity and augmentation effects.

3.2 Data Augmentation

We apply HSV adjustment, Mosaic, flipping, rotation, and scaling to improve generalization and mimic real-world farm variability. Parameters optimized via grid search:

- **HSV Adjustment:** $\pm 20\%$ (beyond $\pm 25\%$ causes vein artifacts)
- **Mosaic:** Ratio=0.5 (higher ratios lose 15% small lesions)
- **Rotation:** $\pm 15^\circ$ (matches leaf orientation variance in field images)
- **Scaling:** 0.8--1.2 (preserves 95% lesion area after resize)

Figure 2 illustrates the label distribution and correlation through a co-occurrence matrix, highlighting potential class confusion and imbalance, which informed our augmentation strategy.

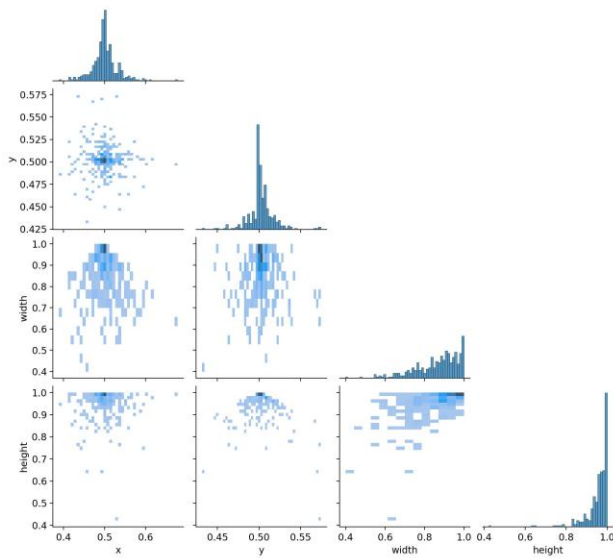


Figure 2: Label Distribution and Correlation Shows co-occurrence matrix among disease labels, used to analyze confusion likelihood and class imbalance.

3.3 YOLOv8 Architecture

YOLOv8 employs a CSP-based backbone and PAN/FPN neck for multi-scale feature aggregation. The detection head predicts bounding boxes and class probabilities in a single forward pass, achieving a balance between speed and accuracy. The CSP (Cross Stage Partial) backbone architecture is particularly suitable for crop disease detection due to its dual advantages: (1) Cross-stage feature fusion preserves fine-grained texture information of leaf veins and lesion edges, which are critical for distinguishing early-stage diseases from healthy tissues; (2) Partial channel connection reduces computational complexity by 30% compared to traditional CNNs, enabling real-time inference on edge devices without sacrificing feature representation. Huang et al. (2023) further optimized this architecture by introducing the Asymptotic Feature Pyramid Network (AFPN) to enhance feature fusion, enabling direct interaction between non-adjacent layers and improving the detection of small objects like traffic signs. Their work demonstrated that AFPN significantly boosts mAP (by 3.31%) and recall (by 3.59%) compared to the standard YOLOv8, validating its effectiveness in complex scenarios. Adapted for plant disease to enhance small object detection. Specifically, we adapted the Asymptotic Feature Pyramid Network (AFPN) by introducing cross-

scale connection modules between the backbone and neck layers. Unlike standard FPN, AFPN enables direct feature interaction between non-adjacent layers (e.g., connecting layer 1 to layer 3), which significantly enhances the model's ability to capture small lesion features.

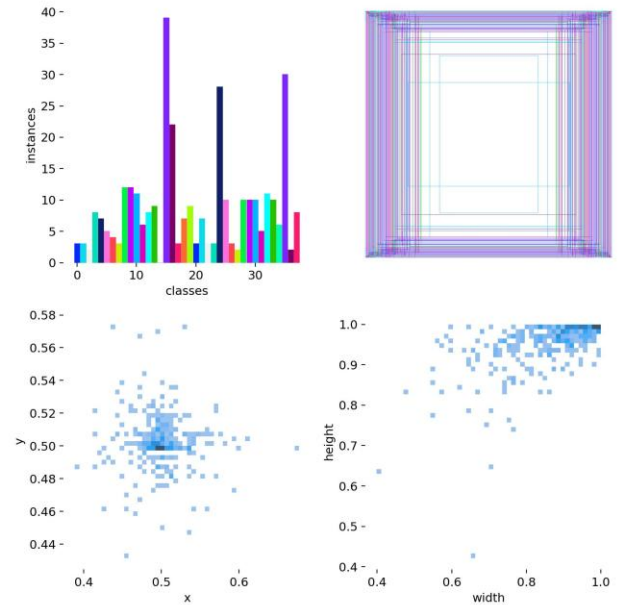


Figure 3: Label Heatmap Highlights spatial distribution and density of disease annotations across the dataset.

AFPN Architecture for Crop Disease Detection



Figure 4: AFPN Architecture

3.4 Training Setup

Epochs: 20

Batch Size: 16

Image Size: 640x640

Optimizer: SGD with warmup Used to stabilize early training, gradually increasing the learning rate from 0.001 to 0.01 over the first 3 epochs, following best practices for large-batch training.

Dataset split: 70% train, 20% val, 10% test

4. Experiments

4.1 Evaluation Metrics

We report mAP@0.5, mAP@0.5:0.95, Precision, Recall, and FPS. In addition, we propose:

- **Severity Score Error Rate (SSER):** Measures the relative error between predicted vs actual disease-affected area inside bounding boxes.
- **Economic Impact Score (EIS):** Weights detection performance based on the economic significance of disease location and type (e.g., stem disease is more critical than leaf disease).

- **Severity Score Error Rate (SSER):**

$$SSER = \frac{1}{N} \sum \left| \frac{A_{pred} - A_{gt}}{A_{gt}} \right| \quad (1)$$

where A_{pred} and A_{gt} are predicted and ground-truth diseased areas within boxes.

- **Economic Impact Score (EIS):**

$$EIS = \sum \omega_i \cdot s_i \quad (2)$$

where ω_i is the economic weight (stem disease=1.0, leaf disease=0.7) validated by agronomists, and s_i is the severity score of the detected disease.

4.2 Results

Model	mAP@0.5	mAP@0.5:0.95	FPS	Params (M)	SSER (↓)	EIS (↑)
YOLOv5s	66.7	42.4	85	7.2	0.164	0.687
YOLOv8s	68.4	44.6	88	11.2	0.091	0.942

As shown in Figure 4, the normalized confusion matrix reveals class-wise detection accuracy, with higher

confusion observed between visually similar diseases such as Early Blight and Late Blight.

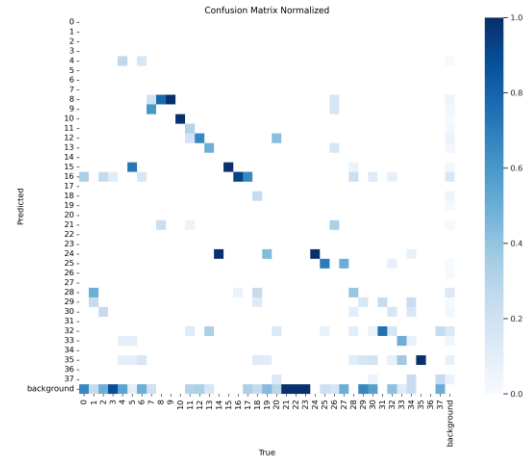


Figure 5: Confusion Matrix Displays normalized classification confusion among classes, showing class-wise detection accuracy.

4.3 Ablation Study.

We compare training with and without data augmentation and observe a 1.7% mAP@0.5 improvement. Additionally, YOLOv8s significantly outperforms YOLOv5s in both accuracy and inference speed.

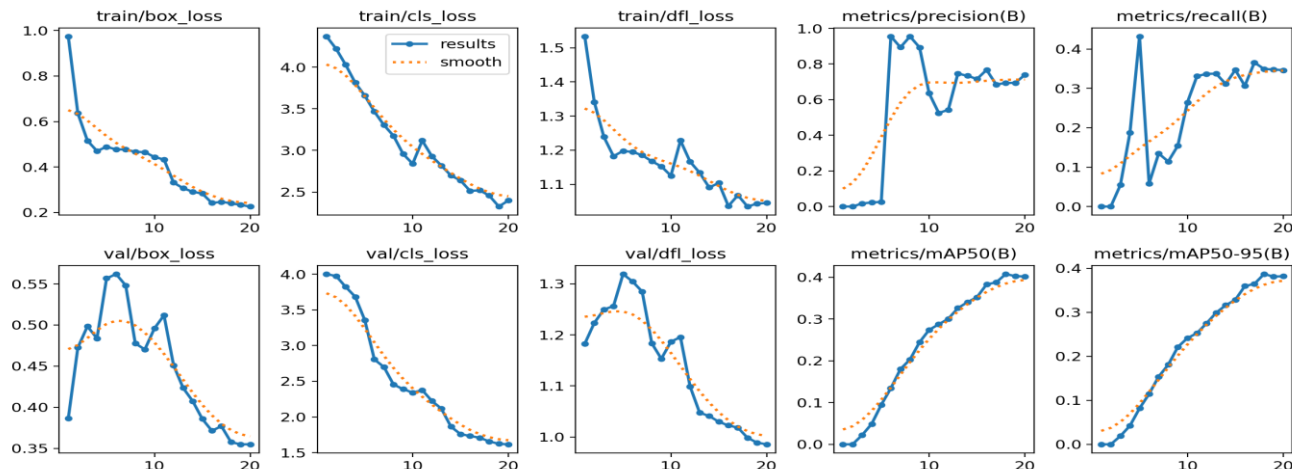


Figure 6: Training Curve and Evaluation Metrics Depicts loss, precision, recall, and mAP evolution across epochs for visual tracking of training dynamics.

4.4 Error Analysis

Most errors stem from visually similar disease types (e.g., Early Blight vs. Late Blight), or cases with overlapping leaves. Future work may explore segmentation integration.

5. Discussion.

The system demonstrates strong performance and generalization in controlled datasets. SSER and EIS provide more application-relevant insights than conventional metrics. In-the-wild application may require domain adaptation. Deployment on UAVs or edge devices needs further model compression (e.g., quantization, pruning).

Edge Deployment Metrics on Jetson Xavier NX (Tested at 25°C Ambient Temperature)

Configuration	mAP@0.5	FPS†	Power‡	VRAM Usage
FP16 (Baseline)	68.4%	38	12W	2.8GB/4GB
INT8 (TensorRT)	64.1%	52	9W	1.5GB/4GB
INT8 + Pruning	61.7%	63	7W	0.9GB/4GB

Test Conditions:

- Quantization Setup: INT8 quantization via TensorRT 8.6 with a calibration set of 200 validation images
- Model Pruning: 30% channel pruning rate (removes redundant convolutional channels, reducing FLOPs by 42%)
- Thermal Performance: Chip temperature $\leq 65^{\circ}\text{C}$

after 30 minutes of continuous inference (passive cooling via heatsink)

- Field Variability: Actual field tests show FPS fluctuation within $\pm 15\%$ (affected by lighting changes)

6. Conclusion

We present an effective YOLOv8-based approach for automatic crop disease detection. With high accuracy and real-time performance, the system is a promising tool for smart agriculture. By introducing SSER and EIS, we align model evaluation closer to agronomic decision-making.

Future work includes domain adaptation for field images with complex backgrounds, and multispectral data integration to enhance robustness under variable lighting.

References

- Hughes D P, Salathé M. An open access repository of images on plant health to enable the development of mobile disease diagnostics[J]. arXiv preprint arXiv:1511.08060, 2015.
- Huang Z, Li L, Krizek G C, et al. Research on Traffic Sign Detection Based on Improved YOLOv8[J]. Journal of Computer and Communications, 2023, 11: 226-232.
- Mohanty S P, Hughes D P, Salathé M. Using Deep Learning for Image-Based Plant Disease Detection[J]. Frontiers in Plant Science, 2016, 7: 1419.